NutriSense meal analysis and advice flow

Meal photo and capture context

class band, service day, camera metadata

Input checks and scale anchor

validated context; 60 mm ArUco measured scale or a disclosed prior

SAM 2.1 food mask and SigLIP 2 class scores

41 WED food classes; ranked evidence and configured serving/nutrient mapping

Automatic meal structure detection

3 x 3 grid prompts, up to 9 distinct regions; deduplicate, then partition overlap pixels once

EVIDENCE AND ROUTING

Accepted and complete

class and item regions can support a school score

Uncertain but usable

low class score, detector failure, or component mismatch

Required stage fails

no usable candidate, portion, or nutrient estimate

Unsupported food

outside configured school policy scope

Per-region portions

measured-card geometry, WED mass head, or flagged size prior

Whole-mask estimate

top food mask and leading meal label; tentative totals no compliance verdict

cannot_verify

withhold totals if required portion or nutrient stage cannot produce a value

out_of_scope

no school assessment for unsupported labels

Nutrient sampling

Monte Carlo, 4,000 samples

ESTIMATED

totals shown; no school score

Coverage-aware policy

energy and protein thresholds

Automatic critique and optional follow-up chat

advisory only; successful turns are saved with the browser record

PASS / BORDERLINE / FAIL